Spring 2023 Math 151C Chapter 5 and 6.1 Notes

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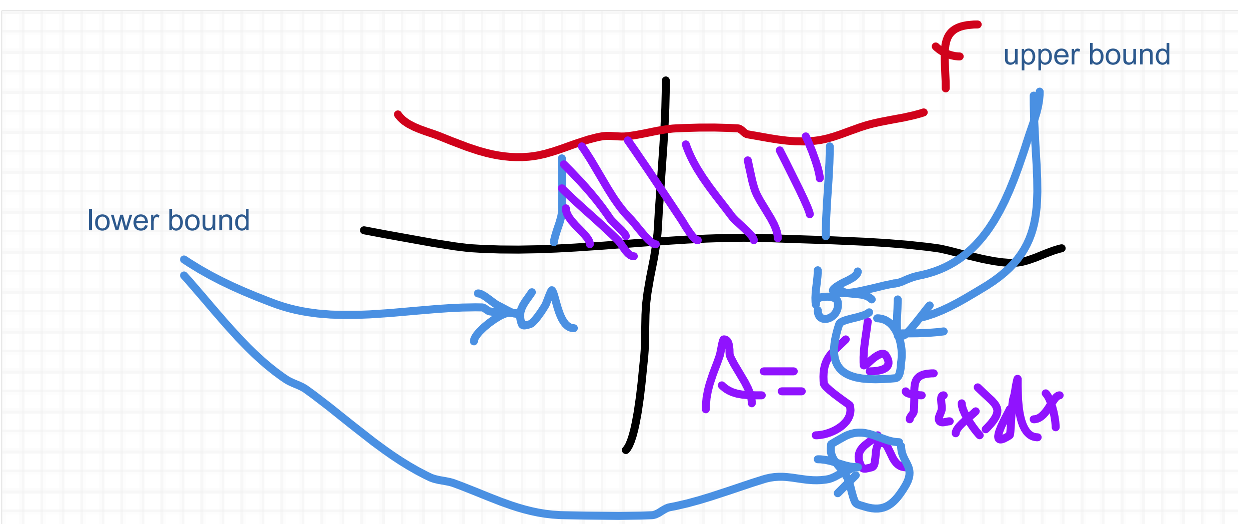
Understanding the Integral

This chapter is about finding the area under the curve. What that entails in theory (as in, by definition) is approximating the area under the curve using objects (such as rectangles) that we understand; in practice (as in, applying mathematical principles), that means using calculus, and what's called the "Fundamental Theorem of Calculus"--and ultimately derivatives (from the perspective of "antiderivatives")--to find the area.

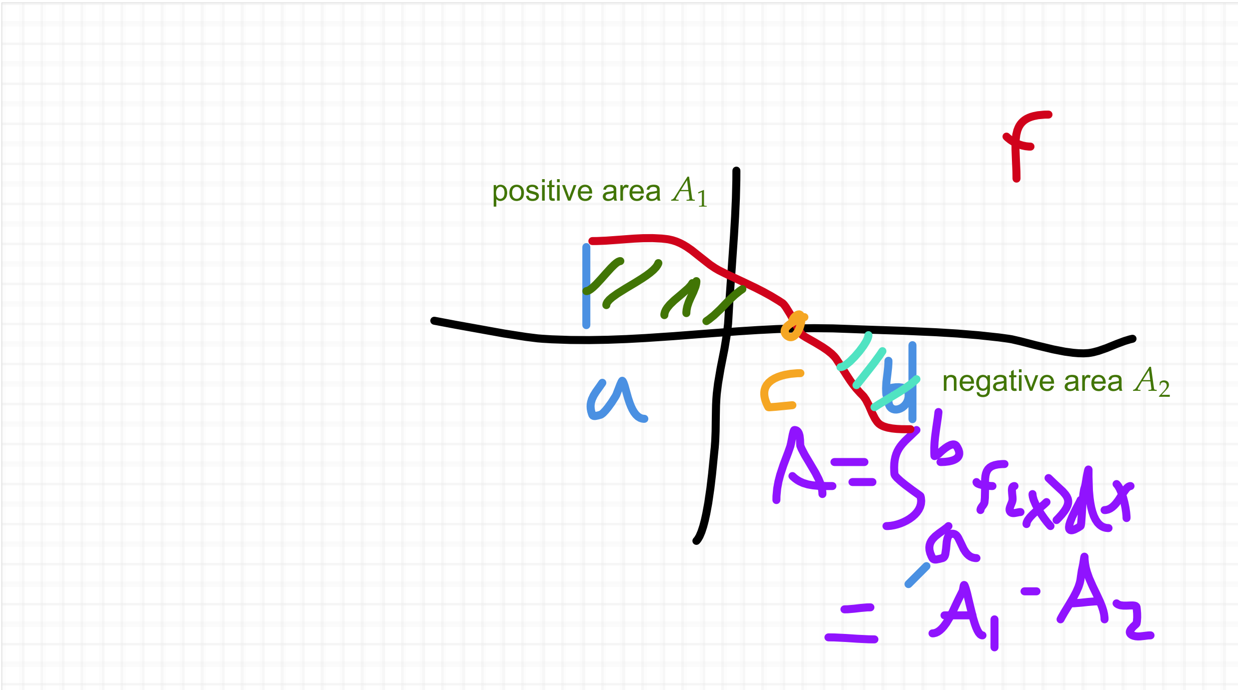
For this course, we're going to quickly discuss the intuitive idea of the "area under a curve", i.e., the definite integral, and then discuss idea of the "antiderivative", i.e. the indefinite integral, and how to find it. Then we relate the two concepts using the fundamental theorem of calculus.

Definition of Definite Integrals

We are interesting in finding the area under an arbitrary curve, i.e., between the curve and the x-axis.



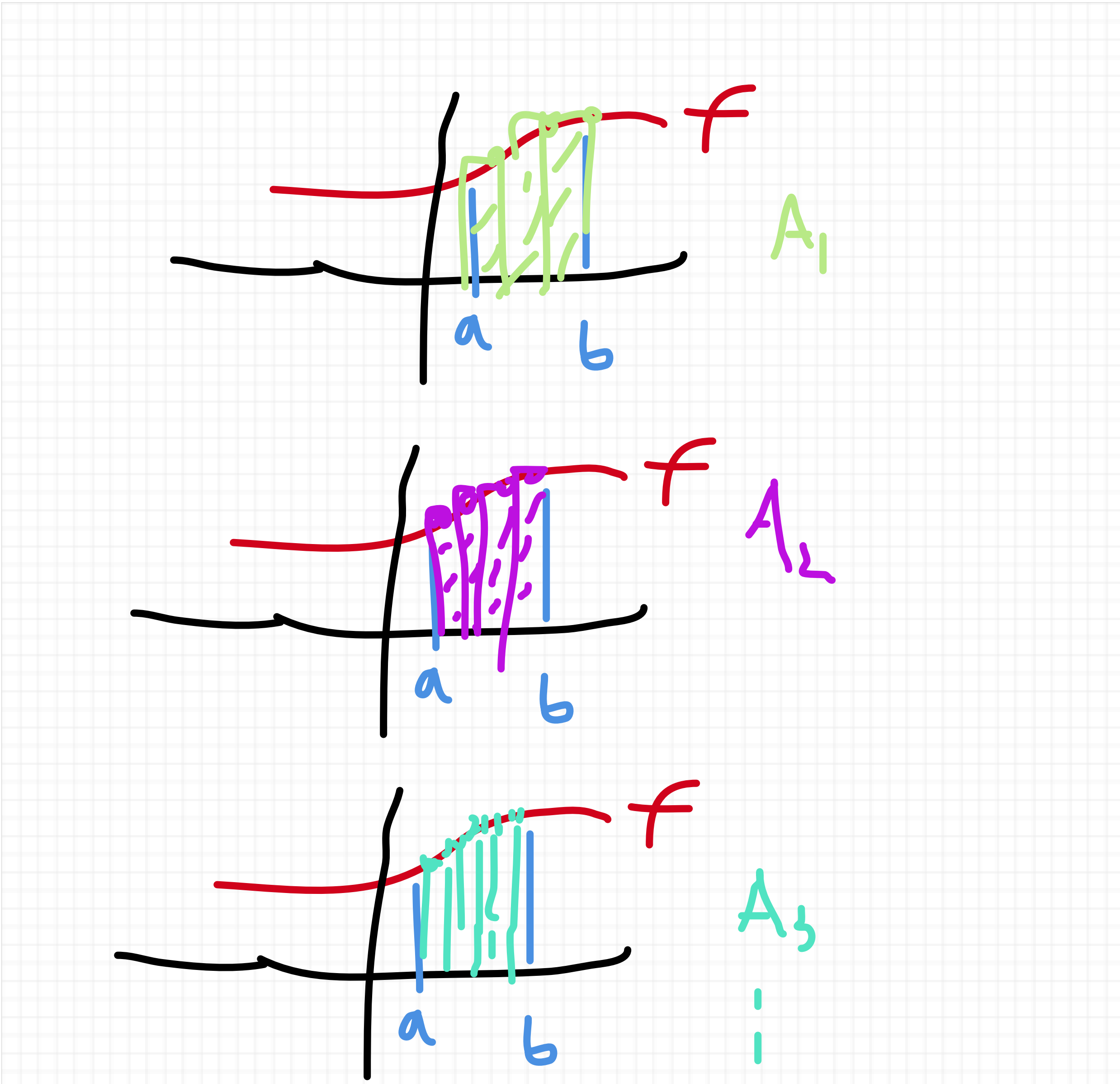
NOTE: What happens if the area we're computing involves the curve going below the x-axis, like in the diagram below



Question: How do we go about actually computing the "area under the curve" in practice?

Answer: We come up with an "equivalent definition" called the "definite integral", and then we use that idea to compute the area.

How one might go about computing the area under an arbitrary curve is approximate the area using geometry, i.e., draw a bunch of shapes that we know how to compute the area of (like a rectangle, triangle, or trapazoid) and then compute the area of finer and finer pieces, until we've found a sufficient approximation



If the function is "well-shaped", then we can approximate the area under the curve in this way using more and more precise (i.e. more rectangles between smaller intervals) and get closer and closer to the actual area.

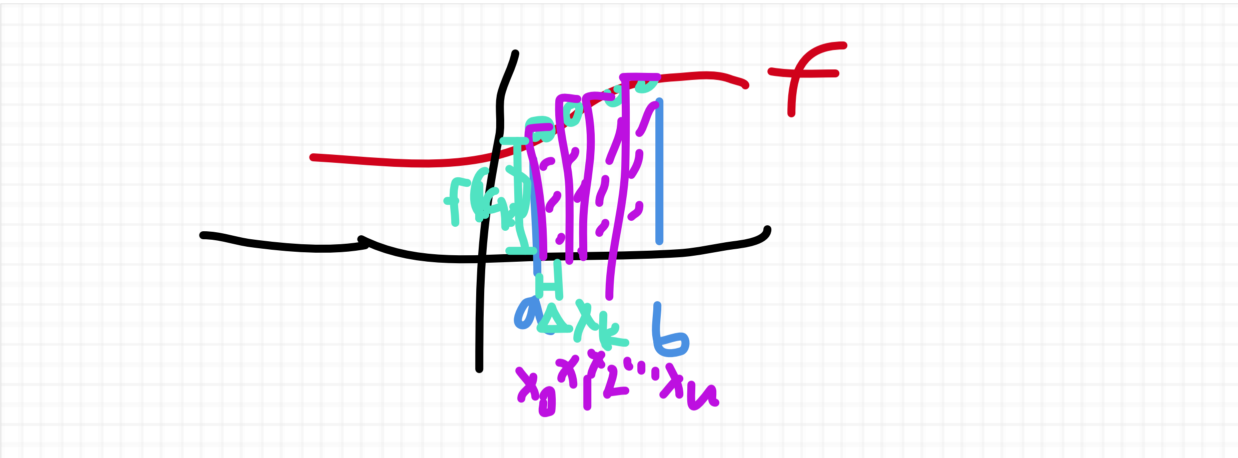
Let's define these ideas more formally.

Sums of these rectangles that compute the area we call **Riemann sums** over an interval $[a, b]$ To specify these Riemann sums, we choose a **partition** $P := \{x_0, x_1, \ldots, x_n\}$, i.e., a set of points $a := x_0 < x_1 < \cdots < x_n := b$

make up a collection of n subintervals $[x_0, x_1], [x_1, x_2], \ldots [x_{n-1}, x_n]$, and a collection of sample points

$C := \{c_1, \ldots, c_n\}$

of c_k in $[x_{k-1}, x_k]$ for $1 \leq c \leq n$ inside each subinterval



So then find the value of the Riemann sum $R(f, P, C)$ of the function f over the partition P with the collection of sample points C by finding the area of the rectangles and adding them all up.

$$R(f, P, C) = \sum_{k=1}^{n} f(c_k)\Delta x_k = f(c_1)\Delta x_1 + \cdots + f(c_n)\Delta x_n$$

where $\Delta x_k := x_k - x_{k-1}$

We define the **mesh** of a partition $||P||$ to be the maximum width of each of the lengths of the subintervals of the partition $[x_{k-1}, x_k]$, i.e. we have

$$||P|| := \max\{\Delta x_k : 1 \leq k \leq n\}$$

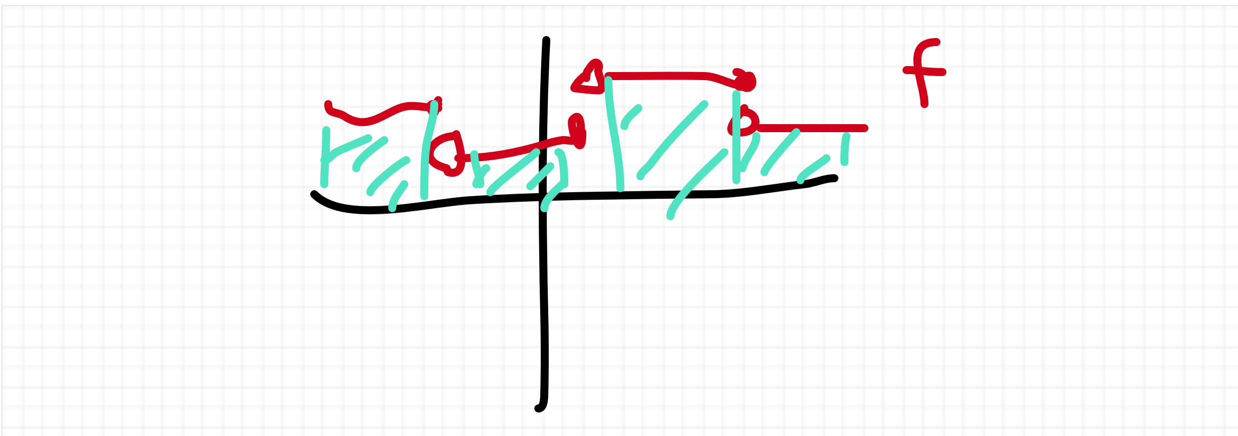
We define the **definite integral** of f over $[a, b]$, denoted by the integral sign (with bounds) $\int_a^b f(x)dx$, as follows:

$$\int_a^b f(x)dx := \lim_{||P|| \to 0} R(f, P, C) = \lim_{||P|| \to 0} \sum_{k=1}^{n} f(c_k)\Delta x_k$$

if such a limit exists.

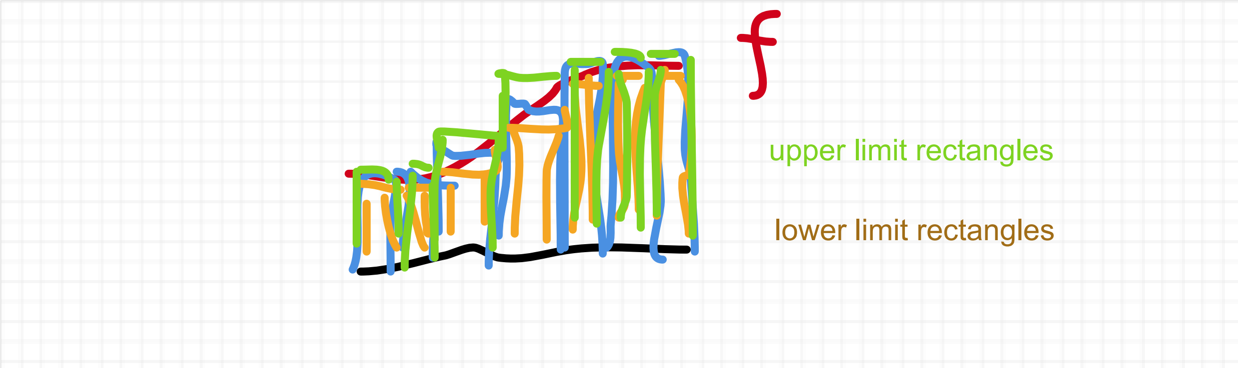
Note that generally for a function f, such a limit may not exist, and computing the area in this way becomes even more difficult, but in general, the types of functions in calculus 1 that we work with do not have this problem.

Theorem 1. *(from Theorem 1 on page 253)* If f is continuous on $[a, b]$, except for finitely many jump discontinuities, in $[a, b]$, then f is integrable over $[a, b]$



Proof. An adequately rigorous proof of this theorem goes way outside the scope of this course, and cannot really be talked about without a background in real analysis. But the basic idea is this: The integral can be found by adding up the integrals of f over the subintervals $I_1, \ldots, I_m$ of $[a, b]$ between the jump discontinuities where f is continuous, and so it suffices to show that f is integrable in the special case where f is continuous over $[a, b]$

Showing that f is integrable if f is continous boils down to showing that what we visually see with approximating the shape of f through a "step function" (which we call a "simple function") approximating the area leads to a well-defined value that is indeed the value we are looking for.



This is usually done by defining an upper limit of Riemann sums, which is defined to be the sum of highest valued-rectangles of subintervals, and a lower limit, which is defined to be the lowest-valued rectangles of subintervals, and using continuity to show that the difference between these upper and lower values get lower and lower as our approximations involve more and more subintervals in our partition (i.e., as $||P|| \to 0$), and ultimately these upper and lower limits agree, leading to existence of the definite integral, as defined above as a limit of when $||P|| \to 0$.

More rigorous details of this proof can be found in Rudin Ch. 5 if you happen to be curious about the "higher-level" mathematics (which to warn you is way more advanced than the Calculus I, let alone Calculus III level) behind this. ◻

Antiderivatives/Indefinite Integrals

Now we're going to talk about a seemingly unrelated concept (found in section 5.3) called the "antiderivative", which also has another name of "the indefinite integral"

An **antiderivative** of a function f on a closed interval $[a, b]$ is a function F such that $F'(x) = f(x)$, or $\frac{d}{dx}[F(x)] = f(x)$

Example 1.

1. *(from example 1 on page 264)* An antiderivative of $f(x) = \cos(x)$ is $F(x) = \sin x$, since

$$F'(x) = \frac{d}{dx}[\sin x] = \cos x = f(x).$$

2. An antiderivative of $f(x) = x$ is $F(x) = \frac{1}{2}x^2 + 2$, since

$$F'(x) = \frac{d}{dx}\left[\frac{1}{2}x^2 + 2\right] = x + 0 = f(x).$$

A natural question one might ask is whether or not the antiderivative is unique. In other words, is say, $F(x) = \sin x$ *the only antiderivative* of $f(x) = \cos x$? The answer to that is not exactly, but we can talk about the general class of antiderivatives, which we shall broadly call "The General Antiderivative" of functions in a fairly nice way.

Theorem 1. *(The Uniqueness of the General Antiderivative)* Let f be a function on $[a, b]$ with an antiderivative F. Then every antiderivative is of the form $G(x) = F(x) + C$, where C is a constant. Another way of saying this is an antiderive is "unique up to a constant".

To prove this theorem, we need the following Lemma:

Lemma 1. Any antiderivative F of $f(x) = 0$ defined on $[a, b]$ is a constant.

Proof. Let F be an antiderivative of f. Given $a < x \leq b$, we find by the **Mean Value Theorem** that there exists $a < c < x$ such that

$$\begin{aligned}
\frac{F(x) - F(a)}{x - a} &= F'(c) = f(c) = 0 \\
\times (x - a) \qquad & \qquad \times (x - a) \\
F(x) - F(a) &= 0 \\
+F(a) \qquad & \qquad + F(a) \\
F(x) &= F(a),
\end{aligned}$$

which shows that $F(x) = C$ for $C = F(a)$ for all x in $[a, b]$. ◻

Proof of Theorem 1. We find that any other antiderivative $G(x)$, we have

$$G(x) = F(x) + [G(x) - F(x)],$$

and we find $C = G(x) - F(x)$ is a constant since

$$\frac{d}{dx}[G(x) - F(x)] = G'(x) - F'(x) = f(x) - f(x) = 0,$$

and that property leads to the conclusion that C is constant by **Lemma 1.** ◻

From **Theorem 1**, we can make sense of the notion of a General Antiderivative of f, which consists of a chosen antiderivative F plus a constant C, which captures the entire family of possible antiderivatives.

We define the **indefinite integral** $\int f(x)dx$ to be the class of all antiderivatives, which we evaluate by finding a single antiderivative F of f and adding an arbitrary constant C to it as so

$$\int f(x)dx = F(x) + C$$

Note that the notation $\int f(x)dx$ of the indefinite integral is like the notation of the definite integral, except it does not contain specific bounds, i.e. the notation for the definite integral of a function f is of the form

We also note that it's not enough to write $\int f(x)dx = F(x)$, since it's a family of all possible antiderivatives, and we must include the constant C. Let's look at the indefinite integrals of the functions discussed in Example 1.

Example 2.

1. We've found in part 1 of Example 1 that for $f(x) = \cos(x)$, we have the antiderivative $F(x) = \sin x$, so we find that

$$\int f(x)dx = \sin x + C \neq \sin x$$

2. We've found in part 1 of Example 1 that for $f(x) = x$, we have the antiderivative $F(x) = \frac{1}{2}x^2 + 2$, so we find that

$$\begin{aligned}\int f(x)dx &= \frac{1}{2}x^2 + (2 + C) = \frac{1}{2}x^2 + C \\ &\neq \frac{1}{2}x^2 + 2\end{aligned}$$

Let's now talk about some important antiderivative rules that will allow you to get a start on the homework

Theorem 2. *(Linearity of Indefinite Integrals, from Theorem 3 on page 265)* Assume that f, g have antiderivatives

Sum Rule:

$$\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx$$

Constant Multiplication Rule:

$$\int cf(x)dx = c\int f(x)dx$$

Proof. Observe that for the antiderivatives F and G of f and g respectively, the result follows immediately from the derivative properties:

$$\frac{d}{dx}[F(x) + G(x)] = F'(x) + G'(x)$$
$$\frac{d}{dx}[cF(x)] = cF'(x) \qquad \square$$

Theorem 3. *(The Inverse Power Rule, from Theorem 2 on page 265)* For $n \neq -1$, we have

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Proof. Follows immediately from the power rule. $\square$

Theorem 4. *(Basic Trigonometric Integrals)* We can work backwards from trig derivatives to get the following antiderivative formulas

$$\int \sin x dx = -\cos x + C$$
$$\int \cos x dx = \sin x + C$$
$$\int \sec^2 x dx = \tan x + C$$
$$\int \csc^2 x dx = -\cos x + C$$
$$\int \sec x \tan x dx = \sec x + C$$
$$\int \csc x \cot x dx = -\csc x + C$$

Proof. Follows from the trig. derivatives. □

Example 3.

1.

$$\int x^2 + 2xdx = \int x^2dx + \int 2xdx = \frac{x^3}{3} + x^2 + C$$

2.

$$\int \sin x + \sqrt{x}dx = \int \sin xdx + \int x^{1/2}dx = -\cos x + \frac{x^{3/2}}{3/2} + C$$
$$= -\cos x + \frac{2}{3}x^{3/2} + C$$

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Example 4. *(from Example 2 in 5.3)*.Using the Linearity rules and the pinverse power rules, we find

$$\int \left(3x^4 - 5x^{2/3} + x^{-3}\right) dx = 3\int x^4dx - 5\int x^{2/3}dx + \int x^{-3}dx$$
$$= 3\frac{x^5}{5} - 5\frac{x^{5/3}}{5/3} + \frac{x^{-2}}{-2} + C$$

Example 5.. *(from Example 3 in 5.3)*

$$\int \left(\sin t + 20\sec^2 t\right) dt = \int \sin tdt + 20\int \sec^2 tdt = -\cos t - 20\sec^2 tdt + C$$

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Fundamental Theorem of Calculus

First Fundamental Theorem of Calculus

Theorem 1. *(The First Fundamental Theorem of Calculus)* Let f be a continuous function on

$[a, b]$. If F is an antiderivative (any antiderivative) of f on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a).$$

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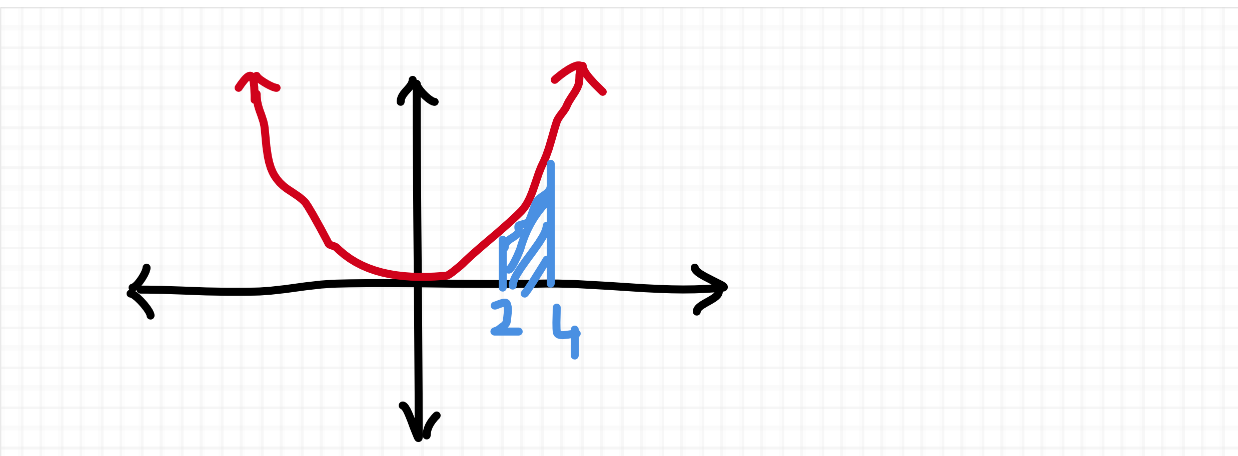
The direct proof of the **First Fundamental Theorem of Calculus** can be found in the book. It is not a particularly "intiutive" or "fulfilling" proof let's just say, and we can use the **Second Fundamental Theorem of Calculus** (which we shall prove) to prove the first as more or less a "Corollary", so we'll skip it.

Example 1. Calculate $\int_2^4 x^2 dx$



Since we find the antiderivative (using the **Inverse Power Rule**) is $\int x^2 dx = \frac{x^3}{3} + C$, so we can use the **First Fundamental Theorem of Calculus** to say that

$$\int_2^4 x^2 dx = \frac{x^3}{3}\bigg|_2^4 = \frac{(4)^3}{3} - \frac{(2)^3}{3} = \frac{64-8}{3} = \frac{56}{3}.$$

Example 2. Calculate $\int_0^1 \sqrt{x} + 5x^4 dx$. We find the antiderivative (using **Linearity** and the **Inverse Power Rule**) is

$$\int \sqrt{x} + 5x^4 dx = \int x^{1/2} dx + 5\int x^4 dx = \frac{x^{3/2}}{3/2} + x^5 + C,$$

and we utilize the **First Fundamental Theorem of Calculus** to conclude that

$$\int_0^1 \sqrt{x} + 5x^4 dx = \left. \frac{x^{3/2}}{3/2} + x^5 \right|_0^1 = \left[\frac{2}{3}(1)^{3/2} + (1)^5\right] - \left[\frac{2}{3}(0)^{3/2} + (0)^5\right] = \frac{5}{3}.$$

Finding the Net Change

As a result of the **First Fundamental Theorem of Calculus**, we can find the "net change" from a rate of change $s'(t)$, i.e., if we're given a rate of change formula s' we can use the **First Fundamental Theorem of Calculus** to find the total change from one point $t = a$ to another point $t = b$ as follows:

total change from a to b $= \displaystyle\int_a^b s'(t)dt = s(b) - s(a).$

Example 1. Water leaks at a formula of $s'(t) = 2 - 5t$ L/h starting at 7pm. What is the total amount of water leaking from 9pm to 11pm?

Solution. To solve this problem, since the water started leading at 7pm, and $t = 2$ hours would have passed between 7pm and 9pm and $t = 4$ hours would have passed between 7pm and 11pm, so we find that

$$\begin{aligned}\text{total amount of water leaking from 7 to 11} &= \int_2^4 s'(t)dt = \int_2^4 2 + 5tdt = \left. 2t + \frac{5}{2}t^2 \right|_2^4 \\ &= \left[2(4) + \frac{5}{2}(4)^2\right] - \left[2(2) + \frac{5}{2}(2)^2\right] \\ &= [8 + 40] - [4 + 10] = 48 - 14 = 34.\end{aligned}$$

Recall that the derivative of displacement $s(t)$ is known (in physics) as velocity $v(t)$.

So if given velocity $v(t)$, we can calculate the total displacement between $t = a$ to $t = b$ by finding

total displacement from a to b $= \displaystyle\int_a^b v(t)dt$

Example 2. if velocity $v(t) = t^3 + 10t^2 - 24t$ compute displacement between $[0, 6]$.

Solution. To find displacement between $[0, 6]$, we find that

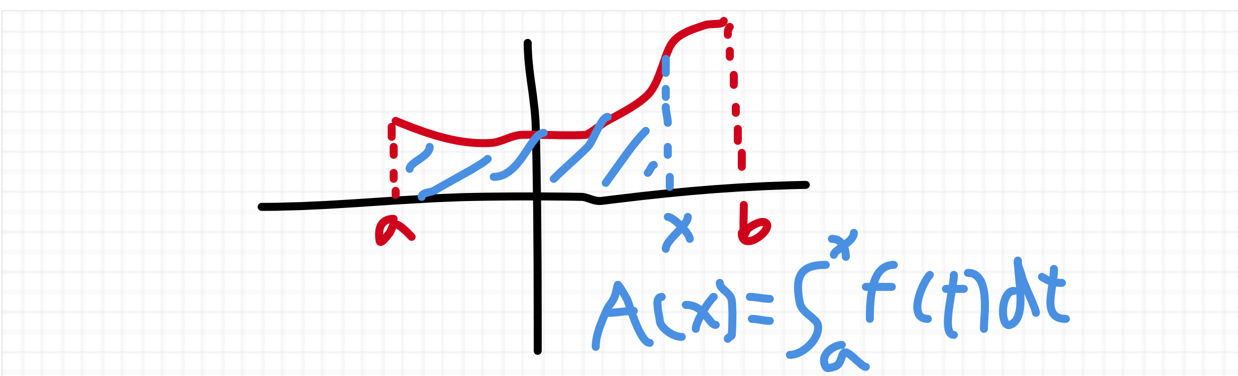
$$\int_0^6 v(t)dt = \int_0^6 t^3 + 10t^2 - 24tdt = \frac{t^4}{4} + \frac{10t^3}{3} - \frac{24t^2}{2}\Bigg|_0^6 = \frac{6^4}{4} + \frac{10(6)^3}{3} - \frac{24(6)^2}{2}$$
$$= 36$$

%FIGURE OUT ERROR WITH THAT

Second Fundamental Theorem of Calculus

Theorem 1. *(Second Fundamental Theorem of Calculus)* Suppose f is continuous on $[a, b]$. (and note by **Theorem 1** of the **Definition of Definite Integrals** section that f is then integrable and hence $\int_a^x f(t)dt$ exists for any x in $[a, b]$). Then the function A defined on $[a, b]$ by $A(x) = \int_a^x f(t)dt$ is the antiderivative of f, i.e., we have

$$\frac{d}{dx}\left[\int_a^x f(t)dt\right] = f(x).$$



Proof.

$$\frac{d}{dx}\left[\int_a^x f(t)dt\right] = \lim_{h \to 0} \frac{\int_a^{x+h} f(t)dt - \int_a^x f(t)dt}{h} = \lim_{h \to 0} \frac{\int_x^{x+h} f(t)dt}{h}$$

Let

$M_{f,h}(x) := \max_{x-h\leq t\leq x+h} f(t),\ m_{f,h}(x) := \min_{x-h\leq t\leq x+h} f(t)$

(Note that these these exist since the **Extreme Value Theorem** tells us that f has an absolute minimum and maximum on the closed interval $[x-h, x+h]$, and since

$m_{f,h}(x) \leq f(t) \leq M_{f,h}(x)$ for all $x-h \leq t \leq x+h$

We find that,

$$m_{f,h}(x)\cdot h = \int_x^{x+h} m_{f,h}(x)dt \leq \int_x^{x+h} f(t)dt \leq \int_x^{x+h} M_{f,h}(x)dt = M_{f,h}(x)\cdot h,$$

and we find since for all x and h, we have

$$m_{f,h}(x) \leq \frac{m_{f,h}(x)\cdot h}{h} \leq \frac{\int_x^{x+h} f(t)dt}{h} \leq \frac{M_{f,h}(x)\cdot h}{h} = M_{f,h}(x),$$

and

$$\lim_{n\to 0} m_{f,h}(x) = \lim_{h\to 0} M_{f,h}(x) = f(x),$$

we find by the squeeze theorem that

$$\lim_{h\to 0}\frac{\int_x^{x+h} f(t)dt}{h} = f(x)\ \square$$

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Example 1. *(From Example 1 on page 279)* Define

$$A(x) = \int_{-\pi}^{x}(1-6t-\cos t)dt$$

We use the first fundamental thoerem of calculus to get that

$$\int_{-\pi}^{x}(1-6t-\cos t)dt = \left.\frac{t}{1}-6\frac{t^2}{2}-\sin t\right|_{-\pi}^{x} = t-3t^2-\sin t\,\Big|_{-\pi}^{x}$$
$$= \left[x-3x^2-\sin x\right]-\left[-\pi-3(-\pi)^2-0\right]$$

$$A(x) = x-3x^2-\sin x+\pi+3\pi^2$$
$$\frac{d}{dx}\left[\int_{-\pi}^{x}(1-6t-\cos t)dt\right] = A'(x) = 1-6x-\sin x$$

Example 2. *(from Exercise 25 on page 283)* asks us to find

$$\frac{d}{dt}\left[\int_{100}^{t}\sec(5x-9)dx\right]$$

Let's set $A(t) = \int_{100}^{t}\sec(5x-9)dx$, and the second fundamental theorem of calculus says that $\sec(5x-9)$ is derivative of A

$$\frac{d}{dt}\left[\int_{100}^{t}\sec(5x-9)dx\right] = \frac{d}{dt}A(t) = \sec(5t-9).$$

A consequence of the second fundamental thoerem of calculus is if we have a function $F(x)$ defined by

$$F(x) = \int_{a}^{g(x)}f(t)dt,$$

and g is differentiable on the relevant intervals, then by the chain rule and the first fundamental thoerem of calculus, we have:

Theorem 2.

$$F'(x) = \left(\frac{d}{du}\left[\int_{a}^{u}f(t)dt\right]\circ g\right)(x)\cdot g'(x)$$
$$= (f\circ g)(x)\cdot g'(x)$$
$$= f(g(x))\cdot g'(x)$$

Example 3. *(from Example 5 of page 281)* Find the derivative of

$$G(x) = \int_{-2}^{x^2} \sin t dt$$

We use the chain rule with the $\int_{-2}^{x} \sin t dt$ as the outter function and x^2 as the inner function, along with the second fundamental thoerem of calculus (on the outter function) to get

$$\frac{d}{dx}\left[\int_{-2}^{x^2} \sin t dt\right] = \sin x^2 \cdot 2x.$$

One final version of the Second Fundamental Theorem of Calculus is the following:

Theorem 3. If h and g are differentiable, then

$$\frac{d}{dx}\left[\int_{g(x)}^{h(x)} f(t)dt\right] = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x), \quad (1)$$

Proof. Recall that $A(x) = \int_{a}^{x} f(t)dt$ is the antiderivative of f, so by the **First Fundamental Theorem of Calculus**, we have

$$\int_{g(x)}^{h(x)} f(t)dt = A(h(x)) - A(g(x)),$$

and then (1) follows from the Chain Rule.

Example 4. *(from Exercise 35 on page 283)* Find the derivative of $\int_{\sqrt{x}}^{x^2} \tan t dt$.

Using (1), we have

$$\frac{d}{dx}\left[\int_{\sqrt{x}}^{x^2} \tan t dt\right] = \tan x^2 \cdot 2x - \tan \sqrt{x} \cdot \frac{1}{2\sqrt{x}}.$$

U-Substitution

Let's say u is differentiable and f has an antiderivative F

$(F \circ u)'(x) = F'(u(x)) \cdot u'(x) = f(u(x)) \cdot u'(x),$

it then follows that for $u = u(x)$, we have:

Theorem 1. *(u-substitution property of antiderivatives)*

$$\int f(u(x)) \cdot u'(x)dx = \int f(u)du = F(u(x)) + C,$$

Proof. Observe that

$$\int f(u(x)) \cdot u'(x)dx = \int (F \circ u)'(x)dx = (F \circ u)(x) + C. \quad \square$$

Think of u substitution as the "Chain Rule in Reverse"
%CONTINUE HERE AND TALK ABOUT HOW BOOK CALLS IT SUBSTITUTION METHOD

In practice, we do u-substitution to evaluate

$$\int g(x)dx,$$

using the following steps:

Step 1. Find u and f, then find u' such that

$$\int g(x)dx = \int f(u(x)) \cdot u'(x)dx$$

Step 2. Apply the substitution of u and divide the inside of the integral by $u'(x)$ and change the "differential" dx to du (one incorrectly can think of this as setting $dx = \frac{du}{u'(x)}$)

to get

$$\int g(x)dx = \int \frac{g(x)}{u'(x)}du = \int \frac{f(u) \cdot u'(x)}{u'(x)}du = \int f(u)du$$

Step 3. Evaluate the integral $\int f(u)du$ by finding the antiderivative F of f with respect to u, and then plug in $u(x)$ as si

$$\int g(x)dx = \int f(u)du = F(u) + C = F(u(x)) + C$$

Example 1. *(from Example 1 on page 292)* Evaluate $\int 3x^2 \sin\left(x^3\right)dx$.

Step 1: Look for an outter function f, inner function u, and something multiplying by f that is a factor of u'. We notice that x^3 is the inner function $\sin\left(x^3\right)$, and $\sin x$ is the outter function, and $\frac{d}{dx}\left[x^3\right] = 3x^2$, hence we can set $f(x) = \sin x$, and $u(x) = x^3$, and we apply u substitution and noting that $u'(x) = 3x^2$, and find that

$$\int 3x^2 \sin\left(x^3\right)dx = \int \frac{3x^2 \sin(u)du}{u'(x)} = \int \frac{3x^2 \sin(u)du}{3x^2} = \int \sin(u)du = -\cos(u) + C$$
$$= -\cos\left(x^3\right) + C.$$

Example 2. *(from Example 2 on page 293)* Evaluate $\int x\left(x^2+9\right)^2dx$. We set $u(x) = x^2 + 9$, which gives us $u'(x) = 2x$, so then we do the substitution as follows:

$$\int x\left(x^2+9\right)^2dx = \int \frac{xu^2}{2x}du = \int \frac{u^2}{2}du = \frac{u^3}{6} + C = \frac{\left(x^2+9\right)^3}{6} + C.$$

So far, we found indefinite integrals using u-substitution. Next, we'll show how to find definite integrals (i.e., integrals of the form $\int_a^b f(x)dx$ rather than $\int f(x)dx$) using u-substitution. We can do this in one of two ways. The first is to just find the antiderivative using u-substitution, then go from

$$\int f(u(x)) \cdot u'(x)dx = F(u) + C = (F \circ u)(x) + C$$

to

$$\int_a^b f(u(x)) \cdot u'(x)dx = (F \circ u)(b) - (F \circ u)(a)$$

For instance:

Example 3. Evaluate

$$\int_0^1 x\left(x^2+9\right)^2 dx.$$

We find from **Example 2** that

$$\int x\left(x^2+9\right)^2 dx = \frac{\left(x^2+9\right)^3}{6} + C,$$

so we find that

$$\int_0^1 x\left(x^2+9\right)^2 dx = \left.\frac{\left(x^2+9\right)^3}{6}\right|_0^1 = \frac{\left(1^2+9\right)^3}{6} - \frac{\left(0^2+9\right)^3}{6} = \frac{10^3-9^3}{6}.$$

Another way to do it is to change the bounds of the integral as we do the u-substitution.

Theorem 2.

$$\int_a^b f(u(x)) \cdot u'(x) dx = \int_{u(a)}^{u(b)} f(u) du = F(u(b)) - F(u(a)).$$

Example 4. *(from Example 7 on page 294)* Evaluate

$$\int_0^2 x^2\sqrt{x^3+1}dx$$

Set $u = x^3 + 1$, and determine that $u' = 3x^2$, and note that $u(0) = 1, u(2) = 9$, and we find that

$$\begin{aligned}\int_0^2 x^2\sqrt{x^3+1}dx &= \int_{u(0)}^{u(2)} \frac{x^2\sqrt{u}}{3x^2} du = \int_1^9 \frac{u^{1/2}}{3} du = \left.\frac{u^{3/2}}{3/2 \cdot 3}\right|_1^9 = \frac{2 \cdot 9^{3/2}}{9} - \frac{2 \cdot 1^{3/2}}{9} \\ &= \frac{2 \cdot 27 - 2}{9} = \frac{52}{9}\end{aligned}$$

Next Time: finish off u substitution really quick, we're going to quickly cover 6.1 (not a big section), and then we'll proceed to the review.

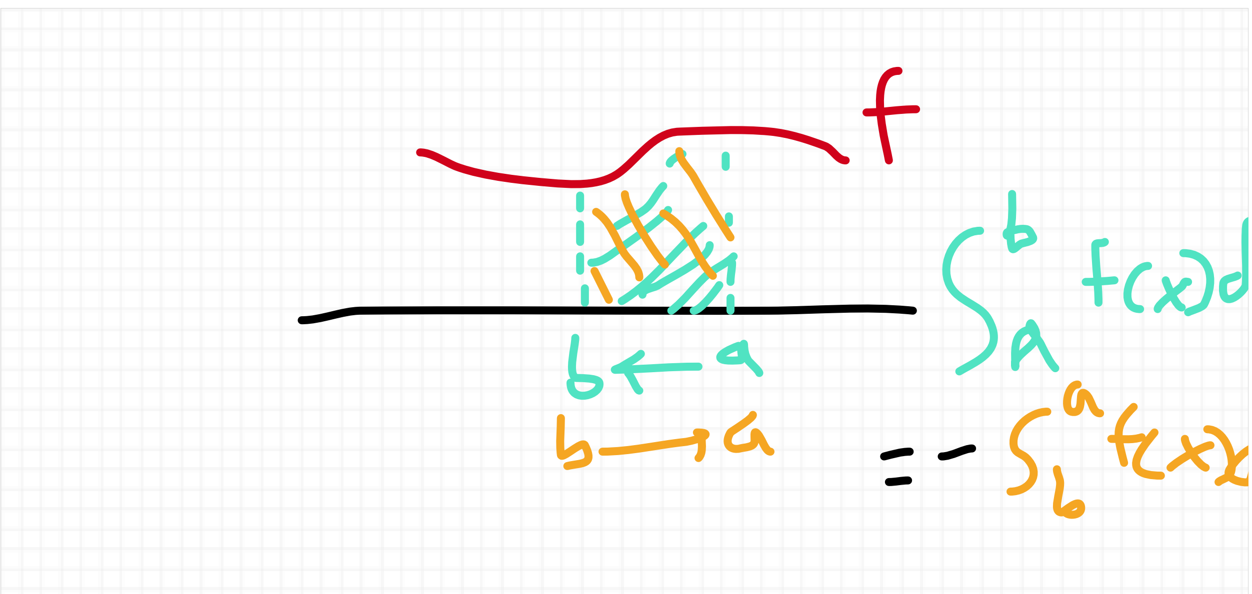
5/11

Something to note is sometimes with u-substitution, you might plug in u and change the bounds so that you have an integral where the "upper bound" b is less than the "lower bound" a.

In this situation (where $a > b$), let's let the **First Fundamental Theorem of Calculus** take its course and we'll see what happens. If F is an antiderivative of f, then

$$\int_a^b f(x)dx = F(b) - F(a) = -[F(a) - F(b)] = -\int_b^a f(x)dx$$

In this situation, we can think of this sort of integral as the negative version of the integral where the greater number a is the upper bound and b is the lower bound



Example 5. Evaluate

$$\int_0^1 \frac{x}{\sqrt{1-x^2}}dx$$

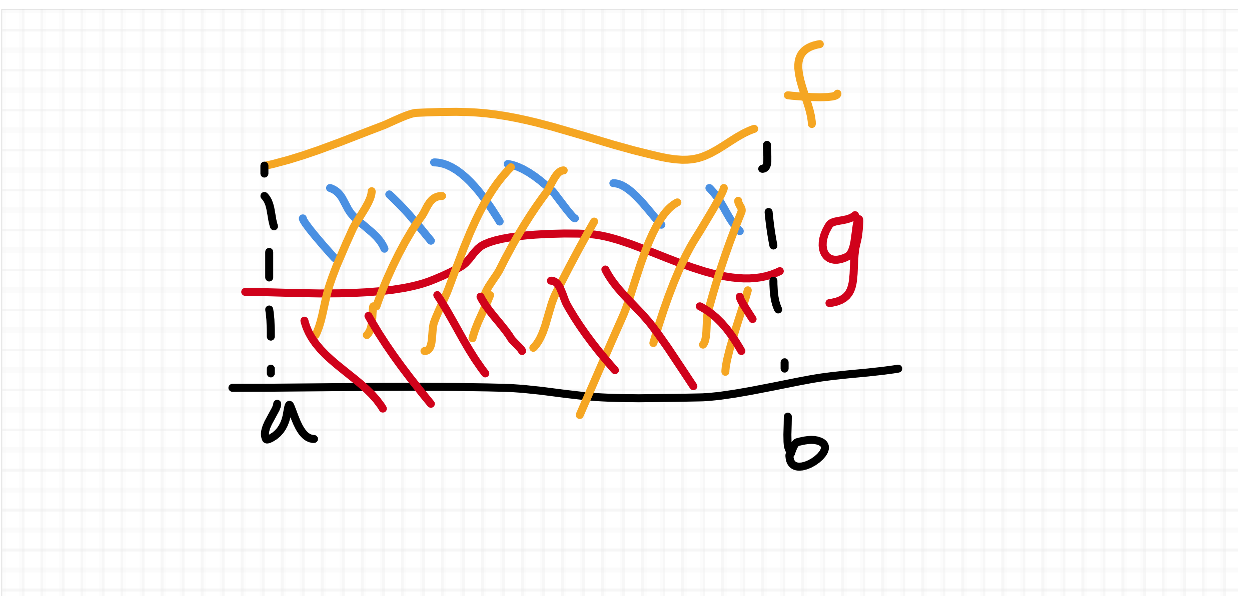
We set $u = 1 - x^2$, and note that $u' = -2x$, and we use u-substitution and we find that

$$\int_0^1 \frac{x}{\sqrt{1-x^2}}dx = \int_{u(0)}^{u(1)} \frac{x}{\sqrt{u}(-2x)}du = \int_1^0 \frac{1}{-2u^{1/2}}du = -\int_0^1 -\frac{u^{-1/2}}{2}du$$
$$= -\left(-\frac{1}{2}\frac{u^{1/2}}{1/2}\bigg|_0^1\right) = \sqrt{1} - \sqrt{0} = 1.$$

Area Between Two Curves

In this section, we learn about finding the area between two curves, of two functions.

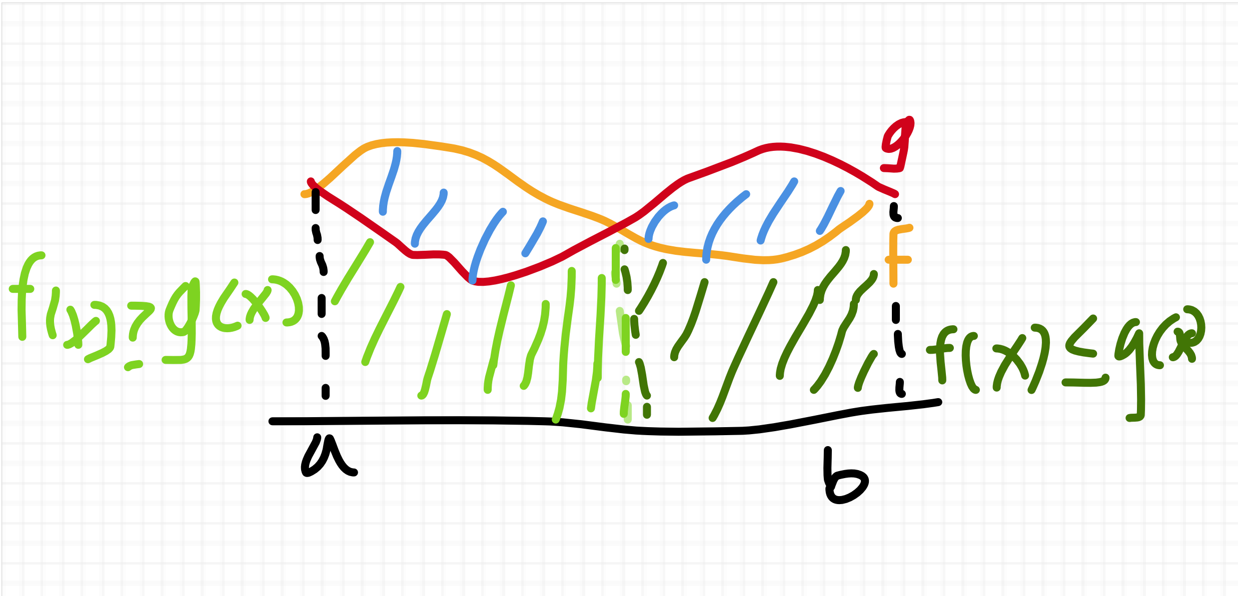
Finding The Area Between Two Curves Over an Interval



In this case, and in cases where f and g are vertically simple in (a, b), i.e., when $f(x) \geq g(x)$ for x in (a, b), we have

$$\text{area between two curves} = \int_a^b f(x)dx - \int_a^b g(x)dx = \int_a^b [f(x) - g(x)]dx$$

One might ask, what if f and g aren't vertically simple on (a, b)?

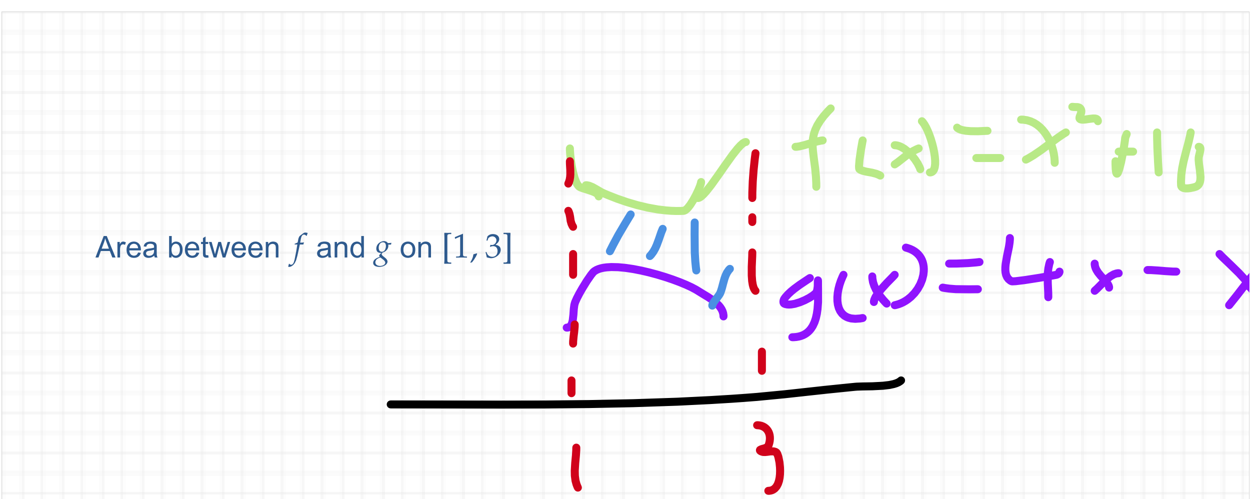


Then we can proceed by using the absolute value of the difference between f and g to account for the different values when $f(x) \geq g(x)$ and when $f(x) \leq g(x)$.

Theorem 1. Given two integrable function

$$\text{area between the curves} = \int_a^b |f(x) - g(x)| dx.$$

Example 1 *(from Example 1 on page 304)* Find the area between the graphs of the functions $f(x) = x^2 + 10,\ g(x) = 4x - x^2$ over the interval $[1, 3]$



Since $f(x) \geq g(x)$, we find

area between the curves $= \int_1^3 |f(x) - g(x)| dx = \int_1^3 f(x) - g(x) dx = \int_1^3 \left[x^2 + 10\right] - \left[4x - x^2\right] dx$

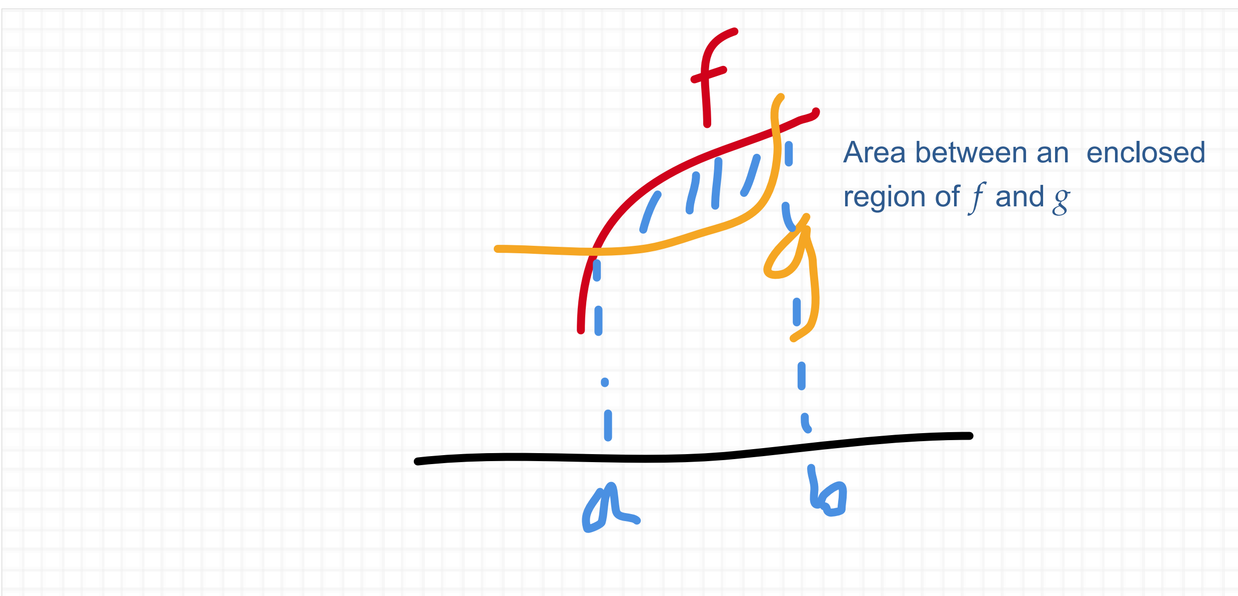
$$= \int_1^3 x^2 + 10 - 4x + x^2 dx = \int_1^3 2x^2 - 4x + 10 dx = \frac{2}{3}x^3 - 2x^2 + 10x \Big|_1^3$$

$$= [18 - 18 + 30] - \left[\frac{2}{3} - 2 + 10\right] = 30 - 8\frac{2}{3} = \frac{64}{3}$$

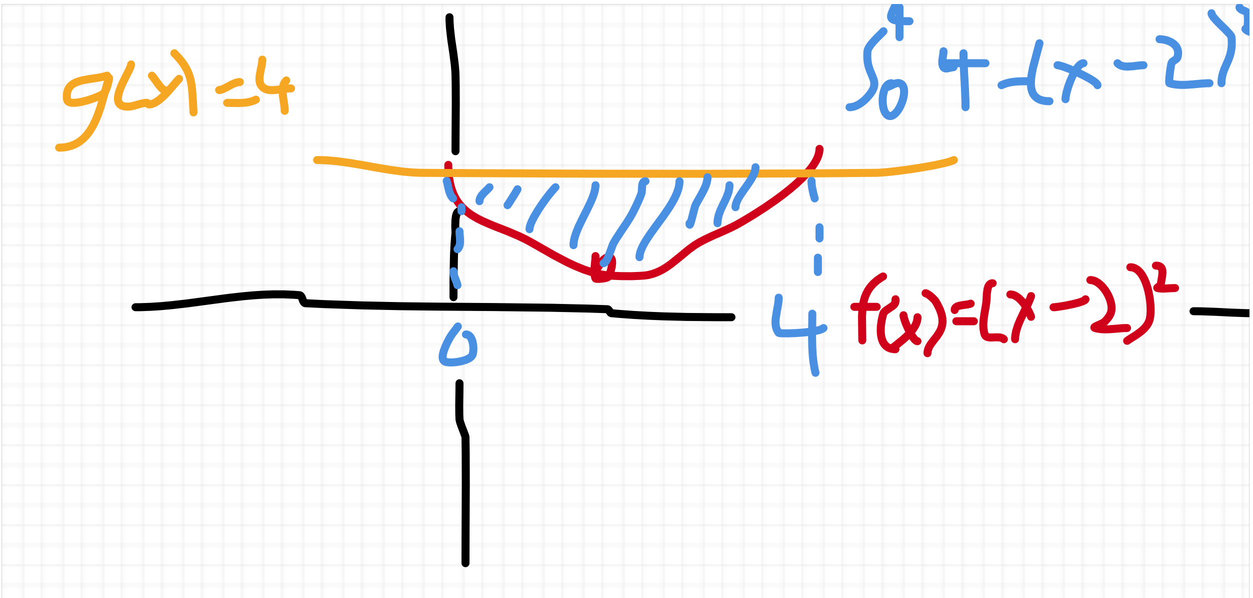
$$= \frac{16}{3}$$

Finding The Area In a Region Between Two Curves

Sometimes we want to find the area between an enclosed region of a few functions



Example 1. Evaluate integral between $f(x) = (x-2)^2$ and $g(x) = 4$



If we draw the picture, we can find the points of intersection, but also find where $f(x) = g(x)$ and solve for x to find the upper and lower bounds

$$
\begin{aligned}
&f(x) = g(x)\\
&(x-2)^2 = 4\\
&x^2 - 4x + 4 = 4\\
&\qquad\quad -4 \quad\;\; -4\\
&x^2 - 4x \qquad = 0\\
&x(x-4) \qquad = 0 \Longrightarrow x = 0, 4
\end{aligned}
$$

Now that we know the upper and lower bounds, we use the formula of the area between two curves as follows:

$$
\int_0^4 |f(x) - g(x)| dx = \int_0^4 g(x) - f(x) dx = \int_0^4 4 - (x-2)^2 dx
$$

$$
\begin{aligned}
u = x-2,\ u' = 1\ &= \int_{u(0)}^{u(4)} \frac{4-u^2}{1} du = \int_{-2}^{2} 4 - u^2 du = 4u - \frac{u^3}{3}\Big|_{-2}^{2} = \left[8 - \frac{8}{3}\right] - \left[-8 - \frac{-8}{3}\right]\\
&= \left[8 - \frac{8}{3}\right] - \left(-\left[8 - \frac{8}{3}\right]\right) = 2 \cdot \frac{16}{3} = \frac{32}{3}.
\end{aligned}
$$

As we see from the above example, we solve this kind of problem in the following steps:

Step 1. Sketch the region/find the points a, b of intersection

Step 2. Set up the integral using the formula of

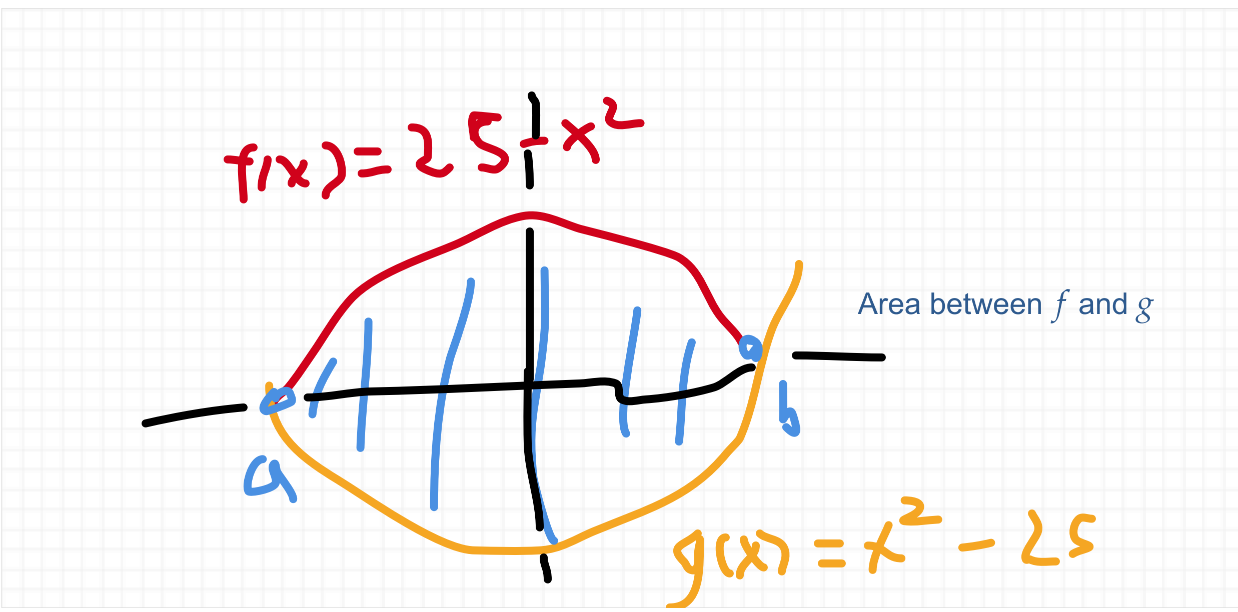
$$\int_a^b |f(x) - g(x)| dx$$

Step 3. Evaluate the integral

Example 2. *(from exercise 31 on page 310)* Find the area in the enclosed region of

$y = 25 - x^2,\ y = x^2 - 25$

Step 1. Sketch the region and find the points of intersection



Based on the sketch, we have $25 - x^2 \geq x^2 - 25$. Then we want to find the points of intersection that enclose the region by solving for x if we set $25 - x^2$ equal to $x^2 - 25$

$$\begin{aligned}
25 - x^2 &= x^2 - 25 \\
+25\ +x^2 &\quad +25 + x^2 \\
50 &= 2x^2 \\
\div 2 &\quad\ \div 2 \\
25 &= x^2
\end{aligned}$$

$$\begin{aligned} \sqrt{25} &= \sqrt{x^2} \\ \pm 5 &= x, \end{aligned}$$

giving us a lower bound of $a = -5$ and an upper bound of $b = 5$

Step 2. Set up the integral

$$\int_{-5}^{5} |[25 - x^2] - [x^2 - 25]| dx = \int_{-5}^{5} [25 - x^2] - [x^2 - 25] dx = \int_{-5}^{5} 50 - 2x^2 dx$$

Step 3. Evaluate the integral

$$\begin{aligned} \int_{-5}^{5} 50 - 2x^2 dx = 50x - \frac{2}{3}x^3 \Big|_{-5}^{5} &= \left[50(5) - \frac{2}{3}(5)^3\right] - \left[50(-5) - \frac{2}{3}(-5)^3\right] \\ &= 2\left[50(5) - \frac{2}{3}(5)^3\right] = 2\left[250 - \frac{2 \cdot 125}{3}\right] \\ &= 2\left[250 - \frac{250}{3}\right] = 2 \cdot \frac{2}{3} \cdot 250 = \frac{1000}{3}. \end{aligned}$$